

Ex 4: Data Modelling and Form Design for a Business Application in Zoho Creator

Topic: Patient Appointment Booking Form

1. Aim

To design and implement a working “Patient Appointment Booking Form” application in Zoho Creator that makes use of every Basic Field offered by the Form Builder, and to apply this design toward a genuine hospital front-desk need — scheduling and pre-registering patient consultation requests.

2. Procedure

2.1 Problem Statement & Application Scope

At present, a hospital receives appointment requests only through phone calls or in-person visits to the front desk. This manual process results in overlapping consultation slots, leaves no advance record of a patient's symptoms, and gives doctors no way to prepare before the patient arrives. Walk-in patients are frequently forced to wait for long periods since there is no prior scheduling mechanism in place.

The objective is to build a low-code Zoho Creator application through which a patient can raise a consultation request using a single online form. This form must record the patient's identity, contact information, referral details, symptoms, preferred schedule, department, and treatment consent — covering all Basic Field types supplied by Zoho Creator's Form Builder.

2.2 Data Model & ER Diagram

The data model comprises an Entity-Relationship (ER) diagram that maps out the schema relationships, along with a Data Dictionary table describing each field's type and constraints. The schema for the Patient_Appointment_Booking_Form application consists of a PATIENT entity holding fixed personal details, an APPOINTMENT entity storing the transactional booking data, and a DEPARTMENT lookup entity that is referenced by the department drop-down field.

2.3 Basic Fields Mapping & Data Dictionary

Table 1 maps each of the thirteen Basic Fields available in the Zoho Creator Form Builder palette to the role it plays in the form, its purpose and validation, and a representative sample value.

Basic Field	Field Link Name	Purpose & Validation Constraints	Sample Value
Name	Patient_Name	Records the patient's full name via First/Last name sub-fields (Mandatory).	Meera Krishnan
Email	Email_Address	Stores a validated email address used to send the appointment confirmation.	meera.krishnan@example.com
Address	Residential_Address	Captures the patient's complete residential address, including city, state and pincode.	22, Nehru Street, Madurai, TN, 625001
Phone	Contact_Number	Stores a validated mobile number the clinic can use to confirm the booking.	+91 98400 11223
Single Line	Referred_By	A short free-text entry naming the referring doctor or source of referral.	Dr. S. Balaji, General Physician
Multi Line	Symptoms_Description	Free-text field describing the patient's current symptoms or reason for the visit.	Persistent headache and mild fever for the past 3 days.

Basic Field	Field Link Name	Purpose & Validation Constraints	Sample Value
Number	Age	Numeric field with validation to record the patient's age.	34
Date	Preferred_Appointment_Date	Date picker used to select the patient's preferred consultation date.	05-Aug-2026
Time	Preferred_Appointment_Time	Captures the patient's preferred time slot for the consultation.	10:30 AM
Drop Down	Department	Single-select list of the hospital's departments or specializations.	General Medicine
Radio	Gender	Single mandatory selection from a set of mutually exclusive options.	Female
Multi Select	Existing_Conditions	Allows one or more pre-existing medical conditions or allergies to be selected.	Diabetes, Allergy to Penicillin
Checkbox	Consent_For_Treatment	Records the patient's consent to proceed with examination and treatment.	Checked (Agreed)

Table 1: Basic Fields – Data Dictionary for the Patient Appointment Booking Form

2.4 Form Layout & Configuration Steps

The Patient_Appointment_Booking_Form was arranged into four logical, patient-friendly sections:

- **Section A — Patient Details:** Patient Name (Name), Age (Number), Gender (Radio).
- **Section B — Contact Details:** Email Address (Email), Contact Number (Phone), Residential Address (Address).
- **Section C — Consultation Details:** Referred By (Single Line), Symptoms Description (Multi Line), Department (Drop Down), Existing Conditions (Multi Select).
- **Section D — Scheduling & Consent:** Preferred Appointment Date (Date), Preferred Appointment Time (Time), Consent For Treatment (Checkbox).

2.5 Workflow Scripting Configuration

Deluge validation scripts were set up under Workflows > Form Workflows > Validations on form submission so that business-logic checks run before a request can be submitted — verifying that the patient's age is valid, that the preferred appointment date is not in the past, and that treatment consent has been given.

3. Output

The figures below show the live preview of the completed form and the submitted patient records as they appear in the report view:

LOW CODE

Home | All Data | Add New | Data Source | Patient_appointment_booking_report

Data Source

Creator tip: Use this sample form to capture data. Below is an example of how it may look.
 Feel free to tweak things via the form builder. Every form submission is recorded in a report and can be visualized.
 Note: If you're capturing sensitive data via a field, take a look at the field's properties to make sure it's secured.

Date: 03-Apr-2026

Patient Name: Vijay P
First Name Last Name

Email Address: vp016@gmail.com

Residential Address: A-5, Vis residential Enclave
Address Line 1
 paravai post
Address Line 2
 Madurai Tamil Nadu
City / District State / Province
 625402 India
Postal Code Country

Contact Number: +91 85258 52900

Referred by: Friend

Symptoms Description: Right shoulder joint pain

Figure 4: Live Application Runtime Preview for Patient_Appointment_Booking_Form Entry (Data Source sample view)

LOW CODE

Home | All Data | Add New | Data Source | Patient_appointment_booking_report

Patient_appointment_booking_report

Date	Patient Name	Email Address	Residential Address	Contact Number	Referred by	Symptoms Description
07-Apr-2026	V Adithya	av0088@gmail.com	d-70, Lalles attan, perumbakkam, Chennai, Tamil Nadu, 600100, India	+918150280105	Dr Arun kumar	ACL (left knee)
03-Apr-2026	Vijay P	vp016@gmail.com	A-5, Vis residential Enclave, paravai post, Madurai, Tamil Nadu, 625402, India	+918525852900	Friend	Right shoulder joint pain
01-Jul-2023						
02-Jul-2023						
03-Jul-2023						
04-Jul-2023						
05-Jul-2023						
06-Jul-2023						
07-Jul-2023						
08-Jul-2023						
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10-Jul-2023						
11-Jul-2023						
12-Jul-2023						
13-Jul-2023						
14-Jul-2023						
15-Jul-2023						
16-Jul-2023						
17-Jul-2023						

Showing 22 of 22

Figure 5: Patient_Appointment_Booking_Form Report View Displaying Submitted Patient Records

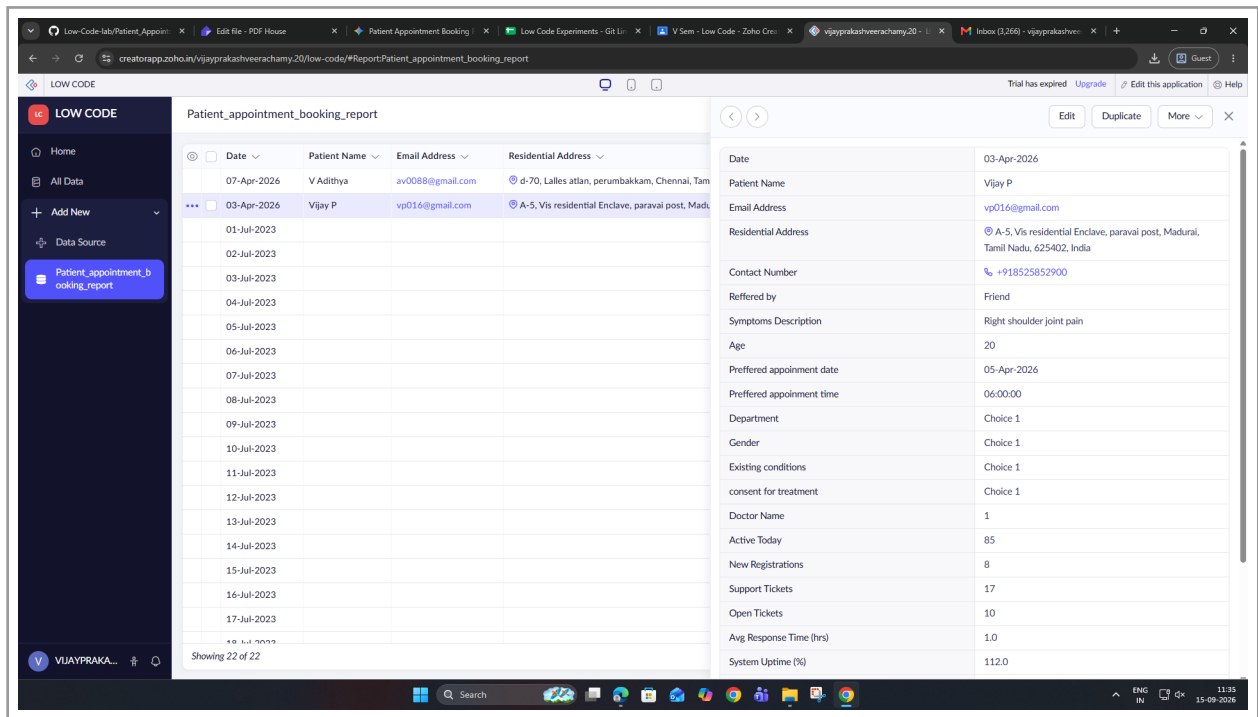


Figure 6: Detailed Record View of a Submitted Patient Appointment Entry

4. Result

The “Patient Appointment Booking Form” application was successfully built in Zoho Creator using all thirteen Basic Fields, complete with field-level validations, sectioned layout formatting, ER schema modelling, and a working report view — fulfilling all the requirements of the assignment.